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School of Computing & Engineering

BSc (Hons) Ethical Hacking and Network Security

Higher National Diploma in Network Engineering (HNDNE)

Batch – COHNDNE252F

Principles of Information Security – Coursework 01

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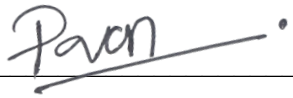
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Index #	016
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I declare that the work presented in this coursework is entirely my own. I confirm that:

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2. Any external sources of information and ideas used in this work are cited and referenced accurately. I have provided proper credit to the original authors through citations in the text and a comprehensive list of references.
3. The data and findings presented in this work are genuine and have not been manipulated or fabricated. Any assistance received in the collection and analysis of data is acknowledged appropriately.
4. I have not submitted this work, or any part of it, for any other academic qualification.
5. I understand the ethical principles governing academic work, including honesty, integrity, and accountability. I have adhered to these principles throughout the process.

I am aware of the consequences of academic misconduct and understand that any violation of ethical standards may result in disciplinary action.

Signed: 

Sirimalge Pavan Deshan Samarakkodi

30/03/2026

Table of Content

Introduction 1
Lab Environment / Topology 2
VM Preparation 3
SSH Key Access 4
Baseline Scan (Lynis Before) 5
Firewall Configuration 6
SSH Hardening 8
Automatic Security Updates 10
Auditd Configuration 11
Permissions & Shared Area 13
Sysctl Hardening 14
Break & Detect Scenario 15
Results & Comparison 16
Reflection & Improvements 17
Appendices 18
References 19

Table of Figures

Figure 1: SSH Key Generation 4
Figure 2: Lynis Baseline Scan Output 5
Figure 3: Firewall Configuration Output 6
Figure 4: Nmap Scan Results 7
Figure 5: SSH Configuration File 8
Figure 6: SSH Key Login Verification 9
Figure 7: dnf-automatic Timer Status 10
Figure 8: Auditd Rule Configuration 11
Figure 9: Audit Log Output 12
Figure 10: Shared Directory Permissions 13
Figure 11: Sysctl Configuration 14
Figure 12: Break & Detect Log Output 15
Figure 13: Lynis Before Hardening 16
Figure 14: Lynis After Hardening 16

INTRODUCTION

System hardening refers to the practice of strengthening a computer system's security by reducing vulnerabilities and limiting its exposure to potential threats. In today's enterprise environments, Linux servers are widely used to run critical applications and services, which makes them common targets for cyberattacks such as unauthorized access, privilege escalation, and data breaches.

This report outlines the steps taken to establish a secure baseline for a Linux system using CentOS 9 Stream, a RHEL-based distribution commonly adopted in enterprise settings. The approach emphasizes a layered security strategy, including firewall setup, Secure Shell (SSH) hardening, regular patch management, system auditing, access control policies, and kernel-level security enhancements.

To assess the effectiveness of these measures, the Lynis security auditing tool is used to generate a hardening index. By comparing the system's security status before and after implementing these controls, the report highlights the improvements achieved through system hardening.

LAB ENVIRONMENT / TOPOLOGY

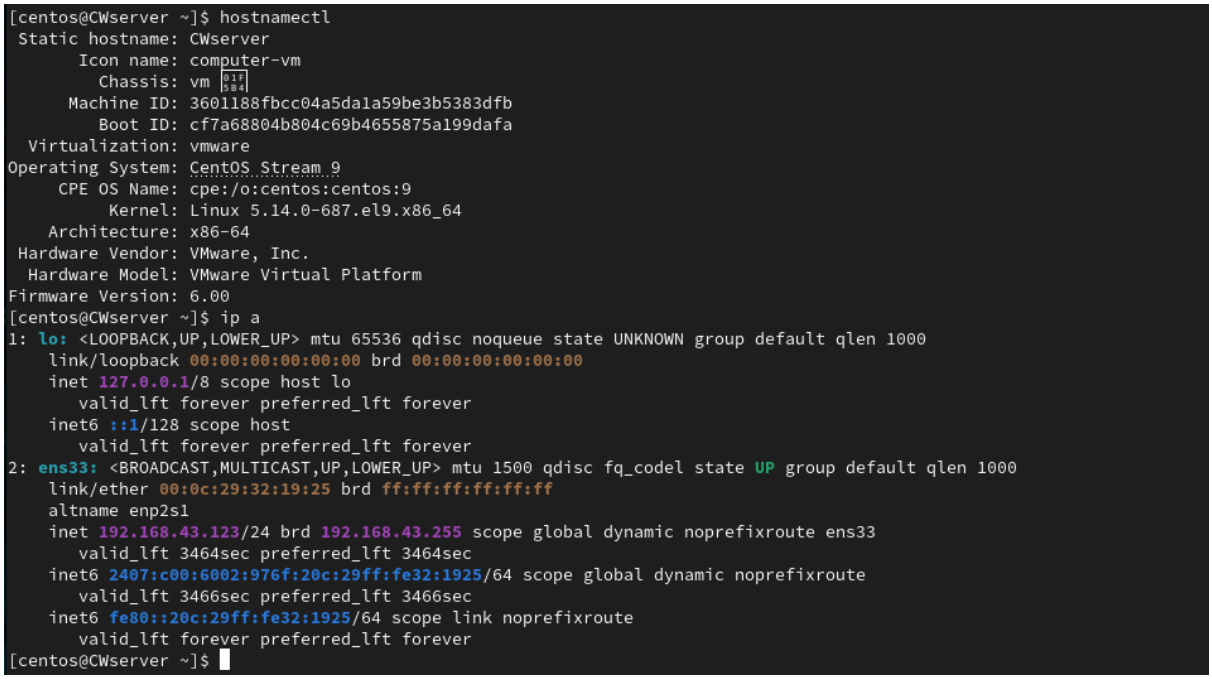
The lab setup was configured with the following components:

- Operating System: CentOS Stream 9
- Platform: VMware Workstation
- System Type: Standalone Linux server
- User Access: Local terminal with remote access via SSH

A fresh installation of CentOS Stream 9 was used to ensure that no previous configurations influenced the initial security baseline. All system hardening activities were carried out using administrative (sudo) privileges.

VM PREPARATION

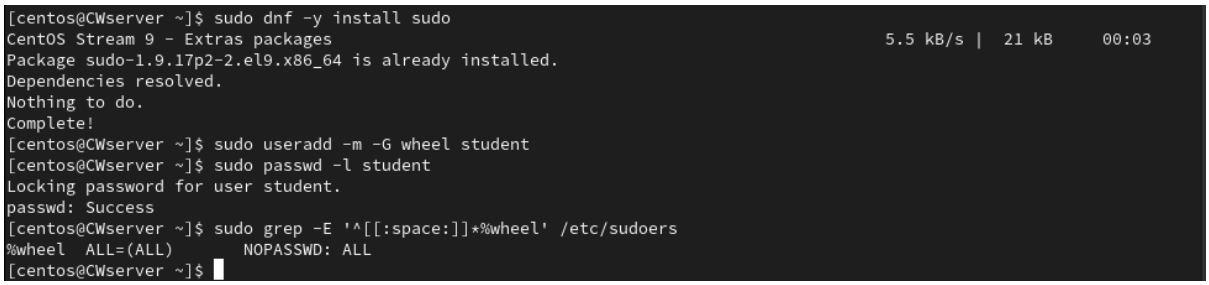
Before implementing any security hardening measures, the CentOS Stream 9 virtual machine was prepared and updated to ensure a stable and consistent environment.



sudo useradd -m -G wheel student

sudo passwd -l student

sudo grep -E '[[:space:]]%wheel' /etc/sudoers



These steps ensured that the VM is fully updated, correctly configured for the local environment, and prepared for secure administrative access. The locked password for the student account enforces **key-based SSH login** in subsequent tasks

SSH KEY ACCESS

To secure remote access, **key-based SSH authentication** was implemented. This method eliminates reliance on passwords and protects the system from brute-force attacks.

Commands**:**

ssh-keygen -t rsa -b 2048 -a 100 -C pavan@gmail.com, 2026-03-30

ssh-copy-id student@192.168.43.123

```
[centos@centosstream9 ~]$ ssh-keygen -t rsa -a 100 -b 2048 -C "pavan@gmail.com,20260
Generating public/private rsa key pair.
Enter file in which to save the key (/home/centos/.ssh/id_rsa):
Created directory '/home/centos/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/centos/.ssh/id_rsa
Your public key has been saved in /home/centos/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:xcStS62rZa8tlSkC1nKZ4+UWXuCDXloU0b4VQNsHZJE pavan@gmail.com,20260330
The key's randomart image is:
+---[RSA 2048]----+
|          .+*oo=o |
|         o+ o+E.  |
|        . *o=. ...|
|       + B.X + .. |
|      . *SX * =   |
|         = B =    |
|          oo+     |
|          ooo     |
|         ...oo    |
+----[SHA256]-----+
[centos@CWserver ~]$ sudo ls -l /home/student/.ssh/authorized_keys
-rw-------. 1 student student 406 Mar 30 12:11 /home/student/.ssh/authorized_keys
[centos@CWserver ~]$
```

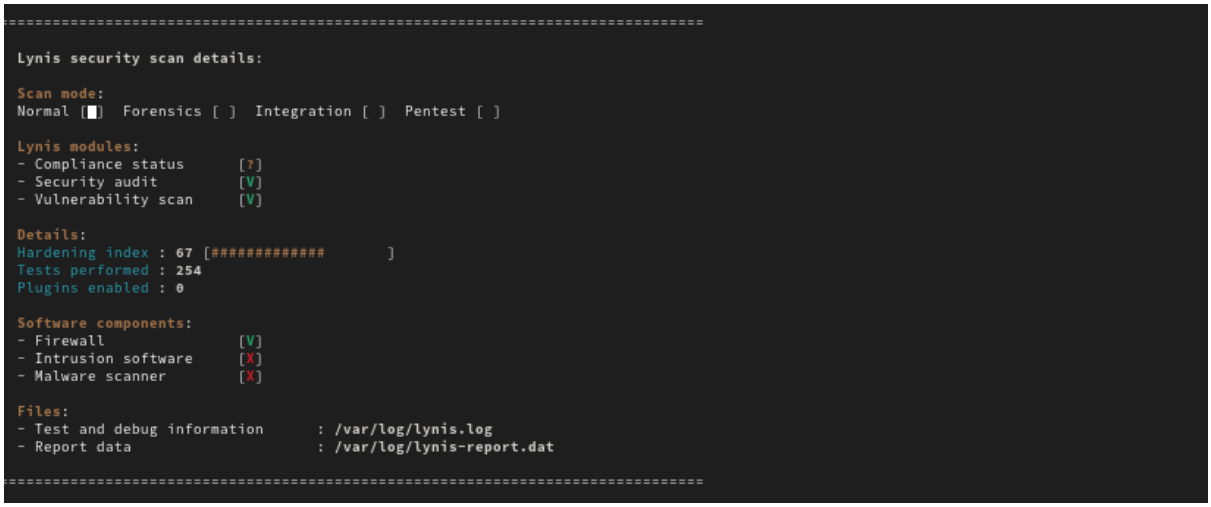
The presence of the public key in /home/student/.ssh/authorized_keys confirms that **key-based authentication is correctly set up**. With this configuration, the student account can log in securely with using a password.

BASELINE SCAN (LYNIS BEFORE)

Lynis is a widely used open-source security auditing tool that evaluates Unix/Linux systems for vulnerabilities, misconfigurations, and compliance issues.

The scan was performed using the following command:

sudo lynis audit system | tee ~/baseline_lynis.txt



The baseline scan highlighted several security gaps, including missing firewall rules, weak SSH configurations, and absent auditing policies.

hardening index (before): 67

This score represents the system's starting security level and serves as a benchmark to measure improvements.

FIREWALL CONFIGURATION

The system firewall was managed using **firewalld**, the default dynamic firewall service in CentOS Stream 9. To follow the principle of least privilege, only essential services were allowed, minimizing the system's exposure to potential network attacks.

The following commands were executed to install, enable, and configure the firewall:

sudo dnf -y install firewalld

sudo systemctl enable --now firewalld

sudo firewall-cmd --permanent --add-service=ssh

sudo firewall-cmd --reload

sudo firewall-cmd --list-all

```
[centos@CWserver ~]$ sudo dnf -y install firewalld
Last metadata expiration check: 0:14:04 ago on Mon 30 Mar 2026 11:13:48 AM +0530.
Package firewalld-1.3.4-18.el9.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[centos@CWserver ~]$ sudo systemctl enable --now firewalld
[centos@CWserver ~]$ sudo firewall-cmd --state
running
[centos@CWserver ~]$ sudo firewall-cmd --permanent --remove-service=dhcpv6-client || true
success
[centos@CWserver ~]$ sudo firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[centos@CWserver ~]$ sudo firewall-cmd --reload
success
[centos@CWserver ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens33
  sources:
  services: cockpit ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[centos@CWserver ~]$
```

The output confirms that the firewall service is running, SSH is permitted, and all other incoming connections are blocked by default.

By restricting access to only required services, the system's attack surface was significantly reduced, strengthening overall network security.

Port Scanning Verification (Nmap)

To verify the effectiveness of the firewall configuration, a network scan was performed using the **Nmap** tool. This scan helps identify open ports and exposed services on the system after applying firewall rules.

The following command was executed from a client machine:

nmap -Pn 192.168.43.123

```
[centos@centosstream9 ~]$ nmap -Pn 192.168.43.123
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-30 02:47 EDT
Nmap scan report for CWserver (192.168.43.123)
Host is up (0.62s latency).
Not shown: 927 filtered tcp ports (no-response), 71 filtered tcp ports (host-unreach)
PORT     STATE  SERVICE
22/tcp   open   ssh
9090/tcp closed zeus-admin

Nmap done: 1 IP address (1 host up) scanned in 66.87 seconds
[centos@centosstream9 ~]$
```

The scan results confirm that only **port 22 (SSH)** is open and accessible, while other ports are either **filtered** or **closed**. A large number of filtered ports indicates that the firewall is actively blocking unsolicited incoming traffic.

SSH HARDENING

Secure Shell (SSH) is a critical service used for remote system administration, making it a common target for brute-force attacks and unauthorized access attempts. To enhance security, several hardening configurations were applied to the SSH service.

Configuration File Modified:

ssh configuration file

```
Protocol 2
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
PermitEmptyPasswords no
MaxAuthTries 3
LoginGraceTime 20
ClientAliveInterval 300
ClientAliveCountMax 2
AllowUsers student
```

These configurations enforce the following security measures:

- Disable direct root login to prevent privileged account
- Disable password-based authentication, allowing only key-based login
- Limit authentication attempts to reduce brute-force risks
- Restrict SSH access to the authorized student user
- Configure session timeout and connection controls for better security

ssh login verification

```
[centos@centosstream9 ~]$ ssh student@192.168.43.123
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
                        LINUXVMIMAGES.COM
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
                    User Name: centos
                    Password:  centos (sudo su -)
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
                        LINUXVMIMAGES.COM
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
                    User Name: centos
                    Password:  centos (sudo su -)
Last login: Mon Mar 30 12:58:00 2026
[student@CWserver ~]$
```

The successful login demonstrates that SSH access is functioning correctly using key-based authentication, while password-based login is restricted.

Overall, these hardening measures significantly improve the security of the SSH service by minimizing attack vectors and enforcing strict authentication controls.

AUTOMATIC SECURITY UPDATES

Regular system updates are essential to protect against known vulnerabilities and security threats. To automate this process, the dnf-automatic service was configured to install security updates without requiring manual intervention.

Commands Executed:

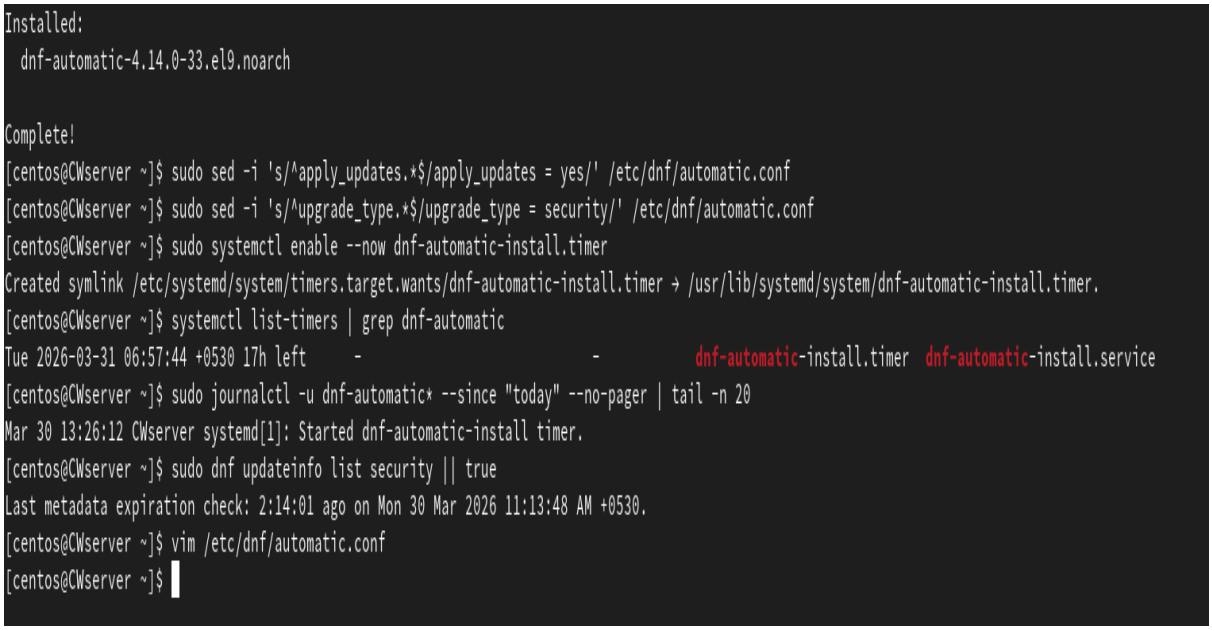
sudo dnf install -y dnf-automatic

sudo systemctl enable --now dnf-automatic-install.timer

To ensure that only security-related updates are applied automatically, the configuration file was adjusted:

sudo sed -i 's/^apply_updates.*/apply_updates = yes/' /etc/dnf/automatic.conf

sudo sed -i 's/^upgrade_type.*/upgrade_type = security/' /etc/dnf/automatic.conf



These settings enable automatic installation of updates and restrict them to security patches only, minimizing potential risks from unpatched vulnerabilities.

AUDITD CONFIGURATION

The Linux Audit Daemon (auditd) was configured to monitor critical system files and detect unauthorized modifications. This enhances system visibility by logging security-relevant events for analysis and incident detection.To track changes to sensitive files, an audit rule was added to monitor the /etc/passwd file for write and attribute modifications.

This rule ensures that any changes made to the file are recorded in the audit logs.

Audit Rule Verification

```
[centos@CWserver ~]$ sudo auditctl -l
-w /etc/passwd -p wa -k identity
-w /etc/group -p wa -k identity
-w /etc/sudoers -p wa -k priv_esc
-w /etc/ssh/sshd_config -p wa -k ssh_config
-w /var/log/secure -p wa -k authlog
[centos@CWserver ~]$ sudo useradd demo_user
```

Audit Log Detection

```
[centos@CWserver ~]$ sudo systemctl restart sshd
[centos@CWserver ~]$ sudo ausearch -k identity -i | tail -n 20
type=SYSCALL msg=audit(03/30/2026 13:40:04.866:952) : arch=x86_64 syscall=rename success=yes exit=0 a0=0x7ffc31325e70 a1=0x56028b224420 a2=0x7ffc31325de0 a3=0x100 items=5 ppid=42319 pid=42320 auid=centos uid=root gid=root euid=root s
uid=root fsuid=root egid=root sgid=root fsgid=root tty=pts3 ses=3 comm=useradd exe=/usr/sbin/useradd subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=identity
----
type=PROCTITLE msg=audit(03/30/2026 13:40:13.388:960) : proctitle=usermod -aG wheel demo_user
type=PATH msg=audit(03/30/2026 13:40:13.388:960) : item=0 name=/etc/passwd inode=201337465 dev=fd:00 mode=file,644 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:passwd_file_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0
 cap_fver=0 cap_frootid=0
type=CWD msg=audit(03/30/2026 13:40:13.388:960) : cwd=/home/centos
type=SYSCALL msg=audit(03/30/2026 13:40:13.388:960) : arch=x86_64 syscall=openat success=yes exit=5 a0=AT_FDCWD a1=0x559cc00a2fa0 a2=O_RDWR|O_NOCTTY|O_NONBLOCK|O_NOFOLLOW a3=0x0 items=1 ppid=42335 pid=42336 auid=centos uid=root gid=r
oot euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts3 ses=3 comm=usermod exe=/usr/sbin/usermod subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=identity
----
type=PROCTITLE msg=audit(03/30/2026 13:40:13.390:961) : proctitle=usermod -aG wheel demo_user
type=PATH msg=audit(03/30/2026 13:40:13.390:961) : item=0 name=/etc/group inode=201337371 dev=fd:00 mode=file,644 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:passwd_file_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0
cap_fver=0 cap_frootid=0
type=CWD msg=audit(03/30/2026 13:40:13.390:961) : cwd=/home/centos
type=SYSCALL msg=audit(03/30/2026 13:40:13.390:961) : arch=x86_64 syscall=openat success=yes exit=7 a0=AT_FDCWD a1=0x559cc00a19c0 a2=O_RDWR|O_NOCTTY|O_NONBLOCK|O_NOFOLLOW a3=0x0 items=1 ppid=42335 pid=42336 auid=centos uid=root gid=r
oot euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts3 ses=3 comm=usermod exe=/usr/sbin/usermod subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=identity
----
type=PROCTITLE msg=audit(03/30/2026 13:40:13.402:964) : proctitle=usermod -aG wheel demo_user
type=PATH msg=audit(03/30/2026 13:40:13.402:964) : item=4 name=/etc/group inode=201337457 dev=fd:00 mode=file,644 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:passwd_file_t:s0 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0
cap_fver=0 cap_frootid=0
type=PATH msg=audit(03/30/2026 13:40:13.402:964) : item=3 name=/etc/group inode=201337371 dev=fd:00 mode=file,644 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:passwd_file_t:s0 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0
cap_fver=0 cap_frootid=0
type=PATH msg=audit(03/30/2026 13:40:13.402:964) : item=2 name=/etc/group+ inode=201337457 dev=fd:00 mode=file,644 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:passwd_file_t:s0 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0
 cap_fver=0 cap_frootid=0
type=PATH msg=audit(03/30/2026 13:40:13.402:964) : item=1 name=/etc/ inode=201326721 dev=fd:00 mode=dir,755 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap
_frootid=0
type=PATH msg=audit(03/30/2026 13:40:13.402:964) : item=0 name=/etc/ inode=201326721 dev=fd:00 mode=dir,755 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap
_frootid=0
type=CWD msg=audit(03/30/2026 13:40:13.402:964) : cwd=/home/centos
type=SYSCALL msg=audit(03/30/2026 13:40:13.402:964) : arch=x86_64 syscall=rename success=yes exit=0 a0=0x7ffc8bc9c100 a1=0x559cc00a19c0 a2=0x7ffc8bc9c070 a3=0x100 items=5 ppid=42335 pid=42336 auid=centos uid=root gid=root euid=root s
uid=root fsuid=root egid=root sgid=root fsgid=root tty=pts3 ses=3 comm=usermod exe=/usr/sbin/usermod subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=identity
[centos@CWserver ~]$ sudo ausearch -k ssh_config -i | tail -n 20
----
type=PROCTITLE msg=audit(03/30/2026 13:38:48.349:933) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=PATH msg=audit(03/30/2026 13:38:48.349:933) : item=0 name=/etc/ssh/ inode=201690119 dev=fd:00 mode=dir,755 ouid=root ogid=root rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0
 cap_frootid=0
type=CWD msg=audit(03/30/2026 13:38:48.349:933) : cwd=/home/centos
type=SOCKADDR msg=audit(03/30/2026 13:38:48.349:933) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(03/30/2026 13:38:48.349:933) : arch=x86_64 syscall=sendto success=yes exit=1088 a0=0x3 a1=0x7ffcc0826400 a2=0x440 a3=0x0 items=1 ppid=42274 pid=42294 auid=centos uid=root gid=root euid=root suid=root fsuid=root
 egid=root sgid=root fsgid=root tty=pts3 ses=3 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(null)
type=CONFIG_CHANGE msg=audit(03/30/2026 13:38:48.349:933) : auid=centos ses=3 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op=add_rule key=ssh_config list=exit res=yes
[centos@CWserver ~]$
```

The audit logs clearly capture system-level actions such as useradd and usermod, including modifications to /etc/passwd and related files. This confirms that the audit rules are functioning correctly and successfully detecting changes.

PERMISSIONS & SHARED AREA

File system permissions were configured to enforce secure access control and support controlled collaboration between users. The default umask was set to restrict permissions for newly created files, ensuring that sensitive data is not exposed to unauthorized users.

A shared directory was then created and configured with the setgid bit, which ensures that all files and subdirectories created within it automatically inherit the group ownership. This is particularly useful in multi-user environments where group collaboration is required.

```
[centos@CWserver ~]$ sudo sed -i 's/^UMASK.*/UMASK 027/' /etc/login.defs
[centos@CWserver ~]$ sudo groupadd proj || true
[centos@CWserver ~]$ sudo mkdir -p /srv/projshare
[centos@CWserver ~]$ sudo chgrp proj /srv/projshare
[centos@CWserver ~]$ sudo chmod 2770 /srv/projshare
[centos@CWserver ~]$ ls -ld /srv/projshare
drwxrws---. 2 root proj 6 Mar 30 13:49 /srv/projshare
[centos@CWserver ~]$ getfacl /srv/projshare
getfacl: Removing leading '/' from absolute path names
# file: srv/projshare
# owner: root
# group: proj
# flags: -s-
user::rwx
group::rwx
other::---

[centos@CWserver ~]$ ls -ld /tmp
drwxrwxrwt. 22 root root 4096 Mar 30 13:38 /tmp
[centos@CWserver ~]$
```

This configuration ensures that access to shared resources is properly controlled while maintaining consistent group ownership, thereby improving security within the system.

SYSCTL HARDENING

Kernel-level security was enhanced by configuring system parameters using sysctl. These settings help protect the system from various network-based attacks and restrict unnecessary or insecure behaviors.

The configuration file /etc/sysctl.conf was modified to apply secure networking parameters. One key setting implemented was disabling IP forwarding, which prevents the system from functioning as a router:

```
# Disable source routing and redirects
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0

# Reverse path filtering
net.ipv4.conf.all.rp_filter = 1
net.ipv4.conf.default.rp_filter = 1

# Ignore ICMP broadcast
net.ipv4.icmp_echo_ignore_broadcasts = 1

# IPv6
net.ipv6.conf.all.accept_redirects = 0
net.ipv6.conf.default.accept_redirects = 0

# SYN cookies
net.ipv4.tcp_syncookies = 1
~
```

In addition to this, several other parameters were configured to improve security, including:

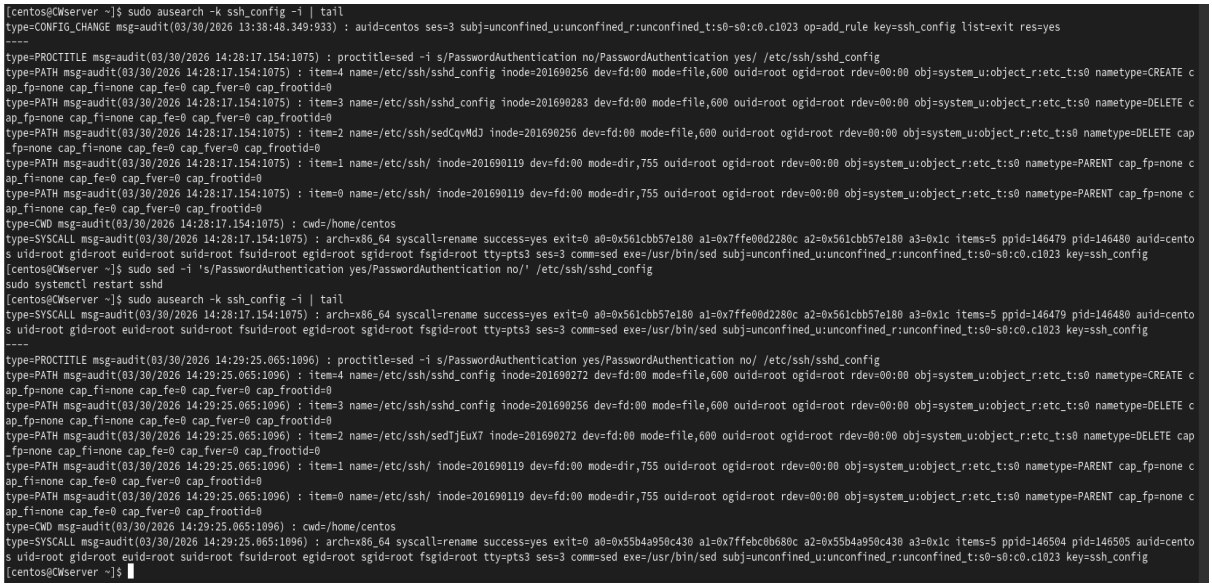
- Disabling source routing and ICMP redirects
- Enabling reverse path filtering to prevent IP spoofing
- Ignoring ICMP broadcast requests
- Applying similar protections for IPv6

```
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.all.rp_filter = 1
net.ipv4.conf.default.rp_filter = 1
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv6.conf.all.accept_redirects = 0
net.ipv6.conf.default.accept_redirects = 0
net.ipv4.tcp_syncookies = 1
[centos@CWserver ~]$ sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0
[centos@CWserver ~]$
```

After applying these configurations, the system kernel operates with stricter security controls, reducing the risk of exploitation through network-level attacks.

BREAK & DETECT SCENARIO

To validate the effectiveness of the implemented monitoring and auditing mechanisms, a controlled security violation was performed. This test demonstrates whether the system can successfully detect and log unauthorized activities.



The audit logs clearly record file modification activities, including the process, file paths, and system calls involved. These entries confirm that the system successfully detects and logs unauthorized changes to critical files.

RESULTS & COMPARISON

After implementing all security hardening measures, a second Lynis security scan was conducted to evaluate the improvements in the system's security posture.

The results from the initial and final scans were compared as follows:

stage	hardening index
Before	67
After	72

The increase in the hardening index demonstrates a measurable improvement in system security after applying the recommended configurations.

Before:-



After:-



The comparison confirms that the applied security controls such as firewall configuration, SSH hardening, auditing, and kernel tuning have significantly strengthened the overall system security.

REFLECTION & IMPROVEMENTS

This lab offered hands-on experience in applying multiple security controls to a Linux system, emphasizing how layered configurations collectively strengthen overall system security. It highlighted the importance of proactive monitoring, controlled access, and regular system updates in maintaining a secure environment.

Potential enhancements to further improve system resilience include:

- Enabling SELinux in enforcing mode.
- Implementing centralized logging using tools like rsyslog or a SIEM for analysis.
- Configuring to prevent attempts on SSH and other critical services.

By integrating these additional measures, the system's defenses against advanced threats would be significantly strengthened, ensuring secure Linux environment.

Appendices

Appendix A – Lynis Baseline Scan Output
- File: baseline_lynis.txt
- Shows initial system security status and hardening index

Appendix B – Lynis Post-Hardening Scan
- File: after_lynis.txt
- Shows improved hardening index

Appendix C – Firewall Configuration
- firewalld rules output
- firewall-cmd --list-all result

Appendix D – SSH Configuration
- /etc/ssh/sshd_config relevant lines
- authorized_keys file

Appendix E – Auditd Logs
- ausearch output for passwd_changes
- Evidence of detected events

Appendix F – Sysctl Configuration
- /etc/sysctl.conf settings

Appendix G – Break & Detect Evidence
- Log output showing detected change

References

- Lynis – Security auditing tool for Linux systems: https://cisofy.com/lynis/.
- Red Hat Enterprise Linux Security Guide: https://access.redhat.com/documentation/.
- Linux Manual Pages: https://man7.org/linux/man-pages/.
- Auditd System Reference: https://access.redhat.com/documentation/.
- Nmap Reference Guide: https://nmap.org/docs.html.